College Physics II
Lecture summaries

DRAFT

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Abstract

Disclaimer: These notes *summarize* the key concepts and formulae discussed during lecture. They should be supplemented by any standard algebra-based physics textbook on these topics (e.g., by Rex and Wolfson) or internet resource to fill in any gaps. Please send any comments, criticisms, suggestions to: `joseph.d.romano@gmail.com`.

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1 Electric charges, forces, and fields

1.1 Electric charge

- Properties of electric charge:

  (i) Two types: *positive* and *negative* (+ and −), whose effects tend to cancel. Opposite charges attract, like charges repel.

  (ii) Total electric charge is *conserved* (cannot be created or destroyed, only transferred from one object to another).

  (iii) Electric charge is *quantized* (integer multiples of the charge of a proton $e = 1.602 \times 10^{-19}$ C; an electron has charge $-e$).

- Materials:

  (i) *Conductors*: one or more electrons per atom are able to move freely through the material. Examples: aluminum, copper, gold, ... (metals).

  (ii) *Insulators*: electrons are bound to atomic nuclei. Examples: rubber, wood, paper, ...

- Some demos:

  1) A plastic rod rubbed with fur is negatively charged; a glass rod rubbed with silk is positively charged. Show that opposite charges attract, like charges repel.

  2) Bits of uncharged paper will be attracted to both negative and positive rods due to charge separation within the atoms / molecules in the paper (an insulator).

  3) A charged rubber ballon sticks to the white board for the same reason as (2).

1.2 Electric force

- *Coulomb's law*: The magnitude of the electric force between two point charges q_1 and q_2 separated by a distance r is

$$F = \frac{k|q_1||q_2|}{r^2} \quad \text{where} \quad k = 8.99 \times 10^9 \ \frac{\text{N} \cdot \text{m}^2}{\text{C}^2} \tag{1}$$

- The force is attractive (directed towards one another) if the two charges have opposite signs; the force is repulsive (directed away from one another) if the two charges have the same signs.

- One sometimes writes

$$k = \frac{1}{4\pi\epsilon_0}, \quad \text{where} \quad \epsilon_0 = 8.85 \times 10^{-12} \ \frac{\text{C}^2}{\text{N} \cdot \text{m}^2} \tag{2}$$

  where ϵ_0 is the so-called *permitivity* of free space (vacuum).

- The electric force on point charge Q due to a set of point charges q_1, q_2, ..., q_N is given by the vector sum of the forces

$$\vec{F}_{Q,\text{net}} = \vec{F}_{1Q} + \vec{F}_{2Q} + \cdots + \vec{F}_{NQ} \tag{3}$$

  where $\vec{F}_{1Q}$ is the electric force that point charge q_1 exerts on Q, etc. This is called the *superposition principle*.

1.3 Electric field

- The electric force between charges is a *non-contact* force. A set of charges creates a *field* to which another charge responds.

- The electric field at a point P in space is a vector $\vec{E}(P)$, whose direction is given by the electric force $\vec{F}_{q_0}(P)$ on a small, positive (so called "test") charge q_0 placed at P, and whose magnitude is given by $F_{q_0}(P)/q_0$:

$$\vec{E}(P) = \vec{F}_{q_0}(P)/q_0 \tag{4}$$

Note that by dividing by q_0, the electric field does not depend on properties of the test charge. It depends only on the charges producing the field.

- For a single point charge q:

$$E(P) = \frac{kq}{r^2} \tag{5}$$

where point P is a distance r from q. The direction of the electric field is radially away from the charge if $q > 0$ and towards the charge if $q < 0$.

- The electric field produced by a set of discrete point charges $q_1, q_2, \cdots, q_N$ is given by the superposition principle:

$$\vec{E}(P) = \vec{E}_1(P) + \vec{E}_2(P) + \cdots + \vec{E}_N(P) \tag{6}$$

where $\vec{E}_1(P)$ is the electric field due to q_1, etc.

1.4 Electric field examples

- Electric dipole: Two charges, $+q$ and $-q$ separated by distance a along the x-axis (located at $x = -a/2$, $x = a/2$, respectively). Then the electric field at point $P = (0, y)$ on the y-axis is given by:

$$E_y(P) = 0\,, \qquad E_x(P) = \frac{kqa}{[y^2 + (a/2)^2]^{3/2}} \tag{7}$$

When $y \gg a$, the electric field has magnitude $E(P) \simeq kqa/y^3$, which falls off as the *cube* of the distance to the dipole.

- The electric field produced by an infinite 2-dimensional sheet of charge with surface charge density σ has magnitude

$$E(P) = \frac{|\sigma|}{2\epsilon_0} \tag{8}$$

The field points perpendicular to the sheet (away from the sheet if $\sigma > 0$; toward the sheet if $\sigma < 0$). The magnitude of the electic field is independent of the distance from the sheet.

- In practice, one doesn't have 2-dimensional sheets that are infinite in extent. But the above results still hold for a finite sheet provided the point P is sufficiently close to the sheet.

- The electric field between two infinite, parallel sheets of charge with surface charge densities $+\sigma$ and $-\sigma$ has magnitude

$$E(P) = \frac{\sigma}{\epsilon_0} \tag{9}$$

The electric field points from the positively-charged sheet to the negatively-charged sheet.

- This last example will be important when we discuss *capacitors* later on.

1.5 Electric field lines

- Pictorial representation of the electric field $\vec{E}$. Direction of field lines point in the direction of $\vec{E}$; density of field lines (number per area perpendicular to $\vec{E}$) is proportional to the magnitude of $\vec{E}$.

- Rules for drawing electric field lines:

  (i) lines begin on + charges, end on − charges (or may extend to infinity);

  (ii) the number of lines leaving or terminating on a charge should be proportional to the magnitude of the charge;

  (iii) electric field lines cannot cross.

- For a point charge, the density of lines at a distance r from the charge is given by $N/4\pi r^2$, which is proportional to the magnitude of the field $E(P) = kq/r^2$ as it should be.

1.6 Conductors in electrostatic fields

- Properties:

  1) $\vec{E} = \vec{0}$ everywhere inside a conductor, for both solid conductors and conductors with *empty* cavities.

  2) If an isolated conductor carries a charge, then that excess charge resides on the outer surface of the conductor.

  3) The electric field just outside the surface of a charged conductor is perpendicular to the surface with magnitude $E = \sigma/\epsilon_0$, where σ is the surface charge density.

  4) For irregularly shaped conductors, the surface charge density σ is greatest where the radius of curvature of the surface is smallest (i.e., where the surface is sharpest).

- Property (1) arises since charges in the conductor rearrange themselves in response to an applied electric field such that the *induced* field associated with these charges exactly cancels the applied field in the conductor, $\vec{E} + \vec{E}_{\rm ind} = \vec{0}$.

- Property (1) implies that any external electric field is shielded from a cavity inside a conductor. That's why you're safe inside a car during a lightning storm.

- Fields within a cavity, produced e.g., by a charge, do make it outside of the conductor. So the outside of a conductor is not shielded from the fields inside a cavity of a conductor.

2 Electrostatic energy, potential, and capacitors

2.1 Review: Work and potential energy

- The *work* done by a force in moving a particle from point A to point B along some path is given by

$$W_{A\to B} = \sum_{A\to B} F_\parallel \Delta s = \sum_{A\to B} (F_x \Delta x + F_y \Delta y + F_z \Delta z) \tag{10}$$

  where Δs is magnitude of the infinitesimal displacement vector $\Delta\vec{s}$ along the path and $F_\parallel$ is the component of the force $\vec{F}$ parallel to $\Delta\vec{s}$. The MKS unit of work is a Joule (1 J $\equiv$ 1 N/m).

- For a general force, the work done depends on the path connecting A and B. If the force is *conservative*, then the work is path independent and depends only on the endpoints.

- Examples of conservative forces are gravity, the spring force, and the electrostatic force. An example of a non-conservative force is friction.

- For conservative forces, one can a associate a *potential energy* U with the work done by the force in moving the particle from A to B:

$$\Delta U \equiv U_B - U_A = -W_{A\to B} = -\sum_{A\to B} (F_x \Delta x + F_y \Delta y + F_z \Delta z) \tag{11}$$

- Note that changing the potential energy U by an additive constant ($U' = U + \text{const}$) does not change the difference in potential energy, since $\Delta U = \Delta U'$. So the zero (or reference point) of the potential energy can be set in whatever way is most convenient for the problem at hand. (We will see that for point charges, the reference point is taken to be infinity.)

- From Equation (11) for ΔU, we see that the components of the conservative force $\vec{F}$ and the potential energy U are related by

$$F_x = -\Delta_x U\,, \qquad F_y = -\Delta_y U\,, \qquad F_z = -\Delta_z U \tag{12}$$

  where $\Delta_x U$ denotes the change in the potential energy U as one moves a distance Δx in the x-direction, and similarly for y and z.

- Hence, we can calculate the force $\vec{F}$ by evaluating how the potential energy U changes in different directions in space.

- Note the minus sign in Equation (11). This means that the potential energy of the system decreases if the force does positive work. For example, as a mass m falls in a gravitational field $\vec{g}$, it loses potential energy, while gaining kinetic energy, $\Delta K = -\Delta U$.

- This last result implies that the total mechanical energy $E \equiv U + K$ of a system is unchanged (i.e., *conserved*) if only conservative forces are acting:

$$\Delta E = \Delta K + \Delta U = 0 \quad \text{(for conservative forces)} \tag{13}$$

2.2 Electrostatic potential energy

- Example: Consider three charges q_1, q_2, q_3 separated by distances r_{12}, r_{13}, r_{23}. The electrostatic potential energy associated with this system of charges can be found by calculating how much work an external agent has to do to "build" the configuration by bringing in charges one-by-one from infinity.

- You first bring q_1 to its position against no field; then q_2 to its position against the field created by q_1; and finally q_3 to its position against the field created by q_1 and q_2:

$$U = \frac{kq_1q_2}{r_{12}} + \frac{kq_1q_3}{r_{13}} + \frac{kq_2q_3}{r_{23}} \tag{14}$$

- The above expression can be extended to an arbitrary number of charges.

2.3 Electrostatic potential

- The above results relating work and potential energy apply as well to the electrostatic force acting on a charge q

$$\Delta U = -W_{A\to B} = -\sum_{A\to B} (F_x\Delta x + F_y\Delta y + F_z\Delta z) = -q\sum_{A\to B} (E_x\Delta x + E_y\Delta y + E_z\Delta z) \tag{15}$$

  where we used $\vec{F} = q\vec{E}$ to obtain the last equality.

- Dividing the above equation by q, we obtain

$$\Delta V \equiv \frac{\Delta U}{q} = -\sum_{A\to B} (E_x\Delta x + E_y\Delta y + E_z\Delta z) \tag{16}$$

  where V is called the *electrostatic potential*.

- The electrostatic potential V is thus the electrostatic potential energy U per unit charge (similar to the electric field $\vec{E}$ being the electrostatic force $\vec{F}$ per unit charge). The MKS unit of electrostatic potential is the volt (V) defined as 1 V = 1 J/C.

- If $\vec{E}$ is everywhere orthogonal to the displacement $\Delta\vec{s}$, then $\Delta V = 0$. This means that the electric field $\vec{E}$ is perpendicular to an *equipotential* surface, which has $\Delta V = 0$.

- Note that the units of $\vec{E}$ can be expressed either as 1 N/C (from $\vec{E} = \vec{F}/q$) or as 1 volt/m (from $E_x = -\Delta_x V$, the change in V in the x-direction). Thus, 1 N/C = 1 volt/m.

- Examples of electrostatic potentials:

  (i) For a uniform electric field $\vec{E}$, we have $\Delta V = -Ed < 0$ if we move in the same direction as $\vec{E}$. Thus, for $q > 0$, the electrostatic potential energy *decreases* as you move in the direction of the field.

  (ii) For a point charge q at the origin, $V(r) = kq/r$, where we have chosen zero potential at $r \to \infty$.

  (iii) For a set of discrete point charges:

$$V(P) = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} + \frac{kq_3}{r_3} + \cdots + \frac{kq_N}{r_N} \tag{17}$$

2.4 Electrostatic potential associated with conductors

- Since $\vec{E} = \vec{0}$ everywhere inside a conductor, $\Delta V = 0$ for any path lying within the conductor. Thus, a conductor is an *equipotential*.

- If a conductor has an empty cavity, then $\vec{E} = \vec{0}$ inside the cavity (and also with the conductor).

  Proof: If that wasn't the case, then one find a path from one side of the cavity to the other having $\Delta V \neq 0$, contradicting that a conductor is an equipotential.

- The surface charge density on the surface of a conductor has its greatest magnitude where the radius of curvature is smallest.

  Proof: Model your conductor as two conducting spheres with radii r_1, r_2 (with $r_1 > r_2$) and charge q_1, q_2 connected by a metal wire. Since this configuration is an equipotential, $kq_1/r_1 = kq_2/r_2$, which implies $q_2/q_1 = r_2/r_1$. It then follows that

$$\frac{\sigma_2}{\sigma_1} = \frac{q_2}{4\pi r_2^2} \Big/ \frac{q_1}{4\pi r_1^2} = \frac{q_2}{q_1}\frac{r_1^2}{r_2^2} = \frac{r_2}{r_1}\frac{r_1^2}{r_2^2} = \frac{r_1}{r_2} > 1 \tag{18}$$

2.5 Capacitors

- A capacitor is a circuit element whose primary function is to *store charge*.

- Given two conductors (of any shape and size) with charges $\pm Q$, respectively, the capacitance C is defined by

$$C \equiv Q/\Delta V \tag{19}$$

  where ΔV is the potential difference between the positively and negatively-charged conductors.

- The value of C is independent of Q and ΔV, depending only on the geometrical arrangement of the conductors.

- Example: A parallel-plate capacitor, having area A and separation d has capacitance $C = \epsilon_0 A/d$.

- One can connect capacitors in *parallel* or *series*, or combinations of the two. Such a combination of capacitors can be reduced to a single *equivalent* capacitance.

- For a parallel combination of C_1 and C_2:

$$C_p = C_1 + C_2 \tag{20}$$

  The proof of this relation uses the fact that the potential differences are equal, $\Delta V_1 = \Delta V_2 = \Delta V$, while the charges add $Q = Q_1 + Q_2$.

- For a series combination of C_1 and C_2:

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} \tag{21}$$

  The proof of this relation uses the fact that the potential differences add, $\Delta V = \Delta V_1 + \Delta V_2$, while the charges are equal, $Q = Q_1 = Q_2$.

- To charge up a capacitor, one needs to do work against the electric field associated with the charge already deposited on the plates.

- If q is the current charge on the capacitor, then the associated potential difference of the plates is q/C. To move additional charge dq against that potential requires work $dW = dq\, q/C$.

- The total work W is given by the area under the potential-difference-vs-charge triangle:

$$W = \frac{1}{2}\frac{Q^2}{C} \tag{22}$$

- This work is the electrostatic potential energy U stored in the capacitor, and it can be written in several equivalent ways:

$$U_E = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}Q\Delta V \tag{23}$$

- A *dielectric* is an insulator placed between the conductors of a capacitor.

- Alignment of polarized molecules within the dielectric creates an induced electric field that *opposes* the applied electric field associated with the free charges $\pm Q$ on the two conductors.

- This induced electric field reduces the total electric field between the conductors, leading to a reduced value of the potential diffence ΔV, and a corresponding increase in $C = Q/\Delta V$.

- The increase in capacitance is encoded in the (dimensionless) dielectric constant $\kappa \geq 1$, defined by $C = \kappa C_0$, where C_0 is the capacitance in the absence of a dielectric.

- For vacuum, $\kappa = 1$; for air, $\kappa = 1.00059$; for paper, $\kappa = 3.7$; for a conductor, $\kappa = \infty$.

3 Electric current, resistance, and circuits

3.1 Current

- Current is the rate of flow of charge through some surface:

$$I = \Delta q/\Delta t \tag{24}$$

  where Δq is the net charge that crosses the surface (area A) in time interval Δt.

- The unit of current is an amp, 1 A $\equiv$ 1 C/s. The direction of current is the direction of flow of the positive charges. Current density J is defined as $J \equiv I/A$.

- For charges to flow in a conductor, we need a non-zero electric field inside the conductor. (We are no longer dealing with electrostatics.) The non-zero field creates a potential difference ΔV between two ends of the conductor.

- Consider a cylindrical conductor with cross-sectional area A, charge carriers with number density n, each with charge q.

- In the absence of an applied electric field, there is no net flow of the charge carriers (e.g., free electrons in the conductor move in random directions with velocity $\vec{v}_{\rm th}$ due to thermal effects).

- In the presence of an applied electric field, the charge carriers accelerate, acquiring a drift velocity v_{drift}, which is several orders of magnitude smaller than the thermal velocity v_{th}. (NOTE: $v_{\text{th}} \sim 10^5$ m/s for electrons at room temperature.)

- Take the length of the cylinder to be $\Delta x = v_{\text{d}}\Delta t$, so that in time interval Δt all of the charges in the cylinder have passed through one end of the cylinder. Then

$$I = \frac{\Delta q}{\Delta t} = \frac{qnA\Delta x}{\Delta t} = qnAv_{\text{d}} \quad \Rightarrow \quad J = qnv_{\text{d}} \tag{25}$$

- Example: Calculate the drift velocity of free electrons in 8-gauge copper wire having diameter $d = 3.26$ mm and current $I = 24$ A. (Note that the number density of free electrons in in copper is $n = 8.47 \times 10^{28}$ m^{-3}.) The answer is $v_{\text{d}} = 2.1 \times 10^{-4}$ m/s, which means that it take more than 1 hr for a free electron to travel 1 meter!

- So it's not the speed of the free electrons that make a light turn on almost instantaneously when you flip a switch. Rather it's the electrostatic force between neighboring electrons which "smooths" out their distribution, so that there isn't an excess or lack of charge carriers in any part of the circuit.

3.2 Resistance

- The drift velocity of charges carriers is produced by a force. If that force is proportional to the applied electric field, then the current density $J = qnv_{\text{d}}$ is proportional to the field, $\vec{J} = \sigma\vec{E}$, where σ is the *conductivity* of the conductor (property of the material).

- Materials for which $\vec{J} = \sigma\vec{E}$ are said to obey *Ohm's law*. (NOTE: Ohm's law really isn't a law; it's an empirical property of certain materials.)

- For a cylindrical piece of conductor with conductivity σ, cross-sectional area A, length l, and applied field $\vec{E}$, we have

$$J = \sigma E \quad \Leftrightarrow \quad I/A = \sigma\Delta V/l \quad \Leftrightarrow \quad R = \rho l/A \tag{26}$$

  where $R \equiv \Delta V/I$ is the *resistance* of object and $\rho \equiv 1/\sigma$ is the *resisitivity* of the material.

- R has units of ohms, 1 $\Omega \equiv 1$ V/A, and ρ has units of $\Omega \cdot$ m. Typical values of ρ are $\sim 10^{-8}$ $\Omega \cdot$ m for metals, $\sim 10^{13}$ $\Omega \cdot$ m for insulators, and ~ 1 to 10^3 for semiconductors.

- Resistivity (and thus resistance) depend on temperature:

$$\rho = \rho_0 \left[1 + \alpha(T - T_0)\right] \tag{27}$$

  where ρ_0 is the resistivity at temperature T_0, and α is the temperature coefficient of (units: 1/°C).

- NOTE: $\alpha \sim 10^{-3}$/°C for most conductors. It is possible for $\alpha < 0$ for semiconductors (increasing T may increase density of charge carriers).

- A *superconductor* is a device whose resistance goes to zero below a *critical temperature* T_{c}, which depends on the material.

- In 1911, Kamerlingh-Onnes discovered that Mercury was superconducting below 4.2 K (recall that 0 K, which is *absolute zero*, corresponds to -273.15 °C). In 1987, the Nobel Prize in Physics was awarded to Bednorz and Müller for discovering superconductivity in copper-oxide materials at relatively high temperatures (~ 100 K). These temperatures are greater than that of liquid nitrogen (77 K), which can be used to cool these high-temperature superconductors.

- Current continues to flow in a superconductor even if the potential difference is removed. No energy is dissipated as heat in a superconducting wire.

- Application: Superconducting magnets made with coils of superconducting wire produce stronger magnetic fields than conventional, non-superconducting, magnets. Superconducting magnets are often used in MRI and NMR machines in hospitals.

3.3 Electrical power and electromotive force

- Recall that power is the time rate of change of energy production or consumption, $P = \Delta E/\Delta t$.
- In an electric circuit, a charge q loses energy $q\Delta V$ as it moves through a circuit from the positive terminal of a battery to the negative terminal. It gains energy $q\Delta V$ as it moves from the negative terminal to the positive terminal *through* the battery. (Chemical reactions provide the energy.)
- Thus, the power delivered to a resistor R with potential difference ΔV is

$$P = q\Delta V/\Delta t = I\Delta V = I^2R = (\Delta V)^2/R \tag{28}$$

- The power delivered to a resistor is dissipated as heat, so-called I^2R *losses*.
- Electromotive force (or EMF) is the potential difference associated with the force per unit charge that drives charges from the negative terminal of a battery (solar cell, electric generator, etc.) to the positive terminal.
- EMF is a potentially misleading term, since EMF is a potential difference, not a force.
- EMF is what's constant in a battery (e.g., a 9-volt transistor battery). EMF is typically denoted by $\mathcal{E}$.
- The output voltage ΔV of a *real* battery depends on the amount of current i flowing out of the battery, according to $\Delta V = \mathcal{E} - ir$, where r is the *internal* resistance of the battery.
- As a battery ages, the internal resistance r increases, which causes ΔV and the power delivered to a load resistor R to become much smaller fractions of $\mathcal{E}$ and the total power produced by the battery, $P_r + P_R$. The increased power delivered to the internal resistance causes the battery to heat up.
- The *maximum* power delivered to a load resistor R occurs when its value agrees with the internal resistance r.

3.4 Series and parallel combinations of resistors

- For a series combination of resistors R_1 and R_2:

$$R_{\text{eq}} = R_1 + R_2 \tag{29}$$

  The proof of this relation uses the fact that the currents are equal for both resistors, $I_1 = I_2 = I$, while the potential differences add, $\Delta V = \Delta V_1 + \Delta V_2$.

- For a parallel combination of resistors R_1 and R_2:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} \tag{30}$$

  The proof of this relation uses the fact that the currents add, $I = 1_1 + I_2$, while the potential differences are equal $\Delta V = \Delta V_1 = \Delta V_2$. Note that $R_{\text{eq}} < R_{\text{smallest}}$ for a parallel combination.

- Application: Christmas tree lights

  (i) Old-fashioned Christmas tree lights were connected in series, which meant that if one bulb burned out, all the other bulbs would go out (no current to those bulbs).

  (ii) Newer Christmas tree lights were connected in parallel to solve that problem. But a parallel connection draws a lot of current, which makes these lights a fire hazard.

  (iii) "Miniature" Christmas tree lights are wired again in series, but each bulb has an insulating *jumper* that becomes conductive when that bulb burn out. So the remaining bulbs stay lit.

- Kirchhoff's rules are used to analyze circuits that cannot be reduced using simple parallel and series combinations of resistors.

- Kirchoff's rules are a consequence of conservation of charge and conservation of energy:

  (i) Junction rule: The sum of all the currents into a node is zero.

  (ii) Loop rule: The sum of all the potential differences around a closed loop is zero.

- When applying Kirchhoff's rules, it is important to follow certain sign conventions: (i) if the source of a potential difference (e.g., a battery) is traversed from the $-$ to $+$ terminal, ΔV is assigned a positive sign; (ii) if current i flows through resistor R in the same direction that the loop is traversed, the potential difference is assigned a negative sign, $-IR$, a so-called "voltage drop".

3.5 Resistor-capacitor (RC) circuits

- RC circuits contain resistors, capacitors and possibly sources of EMF.

- Charging a capacitor: When a capacitor C is charged through a resistor R by an EMF $\mathcal{E}$, the charge increases exponentially while the current decreases exponentially:

$$q(t) = q_{\max}\left(1 - e^{-t/\tau}\right), \qquad I(t) = I_{\max}\, e^{-t/\tau} \tag{31}$$

  where $q_{\max} = \mathcal{E}C$ and $I_{\max} = \mathcal{E}/R$. The combination $\tau \equiv RC$ in the exponential is called the *RC time constant*, since it has units of time.

- These results follow from applying the loop rule to the RC circuit: $\mathcal{E} - IR - q/C = 0$ and then using calculus to solve for $q(t)$.

- Discharging a capacitor: When a capacitor C is discharged through a resistor R, both the charge on the capacitor and the current through the circuit decrease exponentially:

$$q(t) = q_0\, e^{-t/\tau}, \qquad I(t) = I_0\, e^{-t/\tau} \tag{32}$$

  where $I_0 = q_0/RC$.

- These results follow from applying the loop rule to the RC circuit: $q/C - iR = 0$ and then using calculus to solve for $q(t)$.

3.6 Household wiring and electrical safety

- Some important numbers:

  - The EMF of a wall outlet is 120 volts (AC, but that really doesn't matter for this analyis).

  - 14-gauge household wire typically carries up to ~ 20 A of current. It has a resistance of 2.5 Ω/1000 ft, so 100 feet of that wire has a resistance $R_{\rm w} \sim 0.25\ \Omega$.

  - A household appliance (e.g., hair dryer) has $P \sim 1200$ W implying $I = 10$ A and $R_{\rm load} = 12\ \Omega$.

  - A human being has a resistance $R_h \sim 100,000\ \Omega$ (dry skin), but $R_h \sim 1000\ \Omega$ (wet skin).

  - Summary: $R_{\rm w} \sim 0.1\ \Omega \ll R_{\rm load} \sim 10\ \Omega \ll R_{\rm h} \sim 1000\ \Omega$.

- Current kills, not potential difference. $I = 0.01$ A will give a painful shock; $I = 0.1$ A is potentially lethal. (120 V and $R_{\rm h} \sim 1000\ \Omega$ gives $I \sim 0.1$ A, which can kill you.)

- Modern household wire consists of 3 wires: a "hot" wire encased in black plastic (120 V), a "neutral" wire encased in white plastic (typically grounded, so 0 V), and a "ground" wire, usually bare or encased in green plastic, which should be connected to the Earth (i.e., "ground").

- Circuit breakers typically consist of fuses (older houses) or spring-loaded switches, which either "burn out" or "trip" if too much current (e.g., greater than 20 A) flows in the circuit, thus preventing fires.

- NEVER replace an old-fashioned fuse with a penny or any other piece of metal that won't burn out if too much current is flowing in the circuit.

- Household appliances that have a three-prong plug have the ground wire connected to the chassis. If the appliance is connected to a three-prong outlet and the hot wire accidently touches the chassis of the appliance, then the total resistance is approximately that of the wire, which will cause the circuit breaker to trip, which prevents you from being electrocuted even if you are touching the appliance.

- If the appliance doesn't have a ground wire, and the hot wire accidently touches the chassis, then the total resistance of the circuit will be that of the load. The circuit breaker won't trip since the current is normal (i.e., less than 20 A). But if you touch the chassis, you will have 120 V across you, which will produce 0.1 A of current through you if you have wet skin. This can electrocute you.

4 Magnetic fields and forces

4.1 Basic properties

- Two magnetic poles, N and S, similar to $+$ and $-$ electrical charges. Like poles repel, unlike poles attract; isolated magnetic poles do not exist in nature (unlike isolated electric charges).

- Iron filings are attracted to both poles of a magnet. They become magnetized.

- Iron filings align themselves with magnetic field lines. Magnetic field lines form closed loops (as opposed to starting or terminating at N and S poles).

- The Earth has an intrinsic magnetic field (due to motion of charged molten material in its core) with its South magnetic pole located near Earth's North geographic pole. A compass needle is a small magnet which aligns its North pole with Earth's South magnetic pole (points approximately toward geographic North).

- All magnetic fields are ultimately due to the motion of electric charges (e.g., electron orbital motion and instrinsic spin).

4.2 Magnetic force on a charged particle

- The magnetic force on a charged particle is given by magnitude

$$F_B = |q| v B \sin\theta \tag{33}$$

  where θ is the angle between the velocity $\vec{v}$ of the charged particle and the magnetic field $\vec{B}$, and the direction of $\vec{F}_B$ given by the right-hand rule (RHR).

- Right-hand rule: Curl the fingers of your right hand from $\vec{v}$ to $\vec{B}$. Then your thumb points in the direction of $\vec{B}$. Thus, $\vec{F}_B$ is perpendicular to the plane spanned by $\vec{v}$ and $\vec{B}$.

- If the particle is at rest ($v = 0$) or if $\vec{v}$ is parallel to $\vec{B}$, then $\vec{F}_B = \vec{0}$. Measuring $\vec{F}_B$ for a given $\vec{v}$ determines only the components of $\vec{B}$ perpendicular to $\vec{v}$.

- Since $\vec{F}_B$ is perpendicular to $\vec{v}$, the magnetic force does no work on a charged particle. (It can only change the direction of $\vec{v}$, not its magnitude).

- Units of $\vec{B}$: 1 Tesla $\equiv$ 1 N/A $\cdot$ m. The Gauss defined by 1 Tesla $\equiv 10^4$ Gauss is also sometime used.

- Earth's magnetic field ~ 0.5 Gauss; MRI magnet ~ 1.5 T; superconducting magnet ~ 30 T; magnetic field of a neutron star $\sim 10^8$ T.

4.3 Motion of a charged particle in a uniform magnetic field

- The motion of a charged particle in a uniform magnetic field with $\vec{v}$ perpendicular to $\vec{B}$ is *uniform circular motion* with $v = \text{const}$. The speed, radius, magnetic field, and charge-to-mass ratio of the particle are related by

$$R = \frac{mv}{|q|B} \quad \Leftrightarrow \quad T = \frac{2\pi m}{|q|B} \quad \Leftrightarrow \quad f = \frac{|q|B}{2\pi m} \tag{34}$$

  where T is the period and f is the frequency of the circular motion. f is called the *cyclotron* frequency.

- Note the cyclotron frequency is independent of the speed v and radius R.

- If $\vec{v}$ has a component parallel to $\vec{B}$, then the motion is a helix.

- Applications of the motion of a charged particle in a uniform magnetic field include:

  (i) velocity selector: formed from perpendicular uniform electric and magnetic fields. The velocity of a charged particle is unchanged provided $v = E/B$.

  (ii) mass spectrometer: a velocity selector with uniform magnetic field $\vec{B}_1$ followed by another uniform magnetic field $\vec{B}_2$. It is used to select different particle masses by the radii of their circular motion

$$\frac{|q|}{m} = \frac{E}{B_1 B_2 R} \tag{35}$$

  J.J. Thomson (1897) determined the charge-to-mass ratio of the electron this way.

  (iii) a cyclotron: the first particle accelerator (E.O. Lawrence, M.S. Livingston, 1934) constructed from a uniform magnetic field and two "dees", having alternating potential difference, with frequency given by the cyclotron frequency discussed above. A charged particle is accelerated to larger and larger velocities with correspondingly larger and larger radii.

4.4 Magnetic force (torque) on a current-carrying wire (loop)

- The magnitude of the magnetic force on a current-carrying wire is

$$F_B = ILB \sin\theta \tag{36}$$

  where θ is the angle between the direction of the current $\vec{I}$ and the magnetic field $\vec{B}$. The direction of $\vec{F}_B$ is given by the RHR as before.

- The above result is a simple extension of the magnetic force on a single charged particle q to nAL charged particles moving with drift velocity $\vec{v}_\text{d}$ in a wire (recall $\vec{I} = qn\vec{v}_\text{d}A$).

- Consider a rectangular loop of wire (sides a and b) carrying a current I in a uniform magnetic field $\vec{B}$. Assume to begin with that $\vec{B}$ is parallel to the sides having length b and perpendicular to the area vector $\vec{A}$.

- The area vector $\vec{A}$ has magnitude $A = ab$ and direction determined by the RHR: have the fingers of your right hand curl in the direction of the circulating current, then your thumb points in the direction of $\vec{A}$.

- For this case, the total magnetic force on the wire is zero, but there is a non-zero net torque about an axis perpendicular to $\vec{B}$ (passing through the center of the loop):

$$\tau = IAB = \mu B \tag{37}$$

  where $\vec{\mu} \equiv I\vec{A}$ is the *magnetic dipole moment* of the current loop.

- More generally, $\tau = \mu B \sin\theta$, where θ is the angle between $\vec{A}$ and $\vec{B}$.

- The torque is in a direction to rotate the magnetic dipole moment $\vec{\mu}$ in the direction of the magnetic field $\vec{B}$.

- Application: motors (but need to alternate the direction of the current flow in order to keep the loop turning.)

4.5 Sources of a magnetic field

- Oersted (in 1819) discovered that a current-carrying wire produces a magnetic field that circles around the wire in accordance with the right-hand rule (thumb points in the direction of the current; fingers curl in the direction of the magnetic field).

- Examples:

  - field at a perpendicular distance d away from infinitely-long, straight wire: $B = \mu_0 I/2\pi d$

  - field at the center of a circular loop of radius r: $B = \mu_o I/2r$

  - solenoid (current I, n turns per unit length, radius a): $B = \mu_0 nI$

  The direction of $\vec{B}$ for these examples is given by the right-hand rule as discussed previously.

- The quantity μ_0 in the above expressions for $\vec{B}$ is the *permeability* of free space (i.e., vacuum). It has the approximate value

$$\mu_0 \approx 4\pi \times 10^{-7} \ \frac{\mathrm{T \cdot m}}{\mathrm{A}} \tag{38}$$

  It plays a role similar to ϵ_0 for electric fields.

4.6 Force between two current-carrying wires

- Consider two infinitely-long, straight current-carrying wires (currents I_1, I_2) parallel to one another and separated by perpendicular distance d. Then the force per unit length between the wires is

$$\frac{F}{L} = \frac{\mu_0 \, I_1 I_2}{2\pi d} \tag{39}$$

- The above result can be obtained by finding the magnetic field $\vec{B}_1$ produced by wire 1 at the location of wire 2, and then calculating the force that $\vec{B}_1$ exerts on the current I_2.

- If the currents in the two wires flow in the same direction, then the force is attractive; if the currents flow in opposite directions, then the force is repulsive.

4.7 Magnetic properties of materials

- The magnetic properties of materials are determined by the magnetic dipole moments associated with atomic current loops (orbital or spin angular momentum of electrons). A proper treatment of magnetism at the atomic level requires quantum mechanics.

- Ferromagnetic materials (iron, cobalt, nickel) have strong coupling between neigboring magnetic dipole moments associated with unpaired electron spins forming magnetic *domains*. These domains increase in size and align themselves in the direction of an external magnetic field. The induced magnetization remains even if the external field is removed. For $T > T_{\mathrm{curie}}$ (770 °C for iron), the magnetization is lost due to thermal motion of the atoms.

- Paramagnetic materials have magnetic dipole moments associated with unpaired electron spins that align themselves in the direction of an external magnetic field. The coupling between neighboring dipole moments is weak, so thermal effects easily disrupt the alignment of the spin magnetic moments.

- Diamagnetic materials (water, copper, gold, silver bismuth) have magnetic dipole moments associated with the orbital angular momentum of electrons that change in a direction *opposite* to an external magnetic field. The induced magnetization thus repels the external field. Magnetic levitation is possible with diamagnetic superconductors.